

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 7

Expert systems

20 questions from 20 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/12 Feb/March 2022 · Question 5

[Mark scheme](#)

- 5 Some expert systems use a mixture of forward chaining and backward chaining. Some use forward chaining only.

Describe the process of forward chaining.

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..... [3]

9 Insurance consultants have to make decisions about whether or not to accept a customer's application for an insurance policy. They often use expert systems to help them assess the risk involved.

- (a) Such a system would have a knowledge base, which consists of a database of facts and a rules base.

Describe the other components of this expert system.

[4]

- (b)** Explain how this insurance expert system would be used by the consultant and how it would assess the risk.

[4]

(a) Without describing its components, explain how an expert system would produce these schedules.

[4]

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[8]

- 10** An expert system is being trialled for use as part of a study on animal identification. It has been set the following task.

The goal is to decide the colour of a bird named Hortense.
Two facts are provided:

- she sings
- she is 15 cm tall.

The rule base contains the following three rules:

- IF X is an ostrich – THEN X has a long neck
- IF X sings and X is 15 cm tall – THEN X is a canary
- IF X is a canary – THEN X is yellow

The inference engine has concluded that Hortense is yellow. It has taken the facts as data and used the rules base to decide:

- 1 Hortense is a canary
- 2 Hortense is yellow

- (a)** Identify which type of chaining the inference engine used.

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..... [1]

- (b)** Explain the reasoning the inference engine used.

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- 8 Explain what is meant by the term 'data driven' within the context of expert systems.

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- 5 A bank uses an expert system to help customers with financial planning.
- (a) Describe the use of **four** components of an expert system. Your answers **must** relate to financial planning.

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[4]

(b) Describe the benefits, to the bank, of using an expert system.

[6]

(c) State **three** drawbacks, to the bank, of using an expert system.

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10 Scientists are using an expert system to help them identify species of animals.

Describe **four** benefits to the scientists of using an expert system for this purpose.

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7 Describe the term goal driven as used in expert systems.

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[6]

3 There is a fault with a car engine. An expert system is used to help diagnose the fault.

Describe how the following components of an expert system are used in the process of diagnosing the fault.

(a) User interface

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(b) Inference engine

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4 A knowledge base is one component of an expert system.

(a) Describe what is meant by the term knowledge base.

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(b) Describe how a knowledge base is used in an expert system.

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3 An expert system contains an inference engine and a knowledge base.

(a) Describe how the inference engine performs its role.

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(b) Describe how a knowledge base is created.

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[5]

[8]

7 A mechanic uses an expert system.

This system has a user interface and an explanation system.

(a) Identify and describe **two** other components of an expert system.

component 1

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component 2

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[4]

The expert system is used to diagnose faults in a car's engine.

(b) Evaluate the use of this system.

[8]

10 An expert system is used to help identify different types of flowers.

(a) Describe the role of an inference engine in this system.

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(b) Describe the role of a knowledge base in this system.

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..... [2]

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 5

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Question	Answer	Marks
5	Three from: An inference engine using forward chaining searches the inference rules... ...until it finds one where the IF statement is known to be true When such a rule is found, the inference engine uses the 'THEN' part to cause the addition of new information/facts The inference engine will repeat this process until a goal is reached The data entered determines which rules are selected and used and so this method is called data-driven	3

Mark scheme · 9626/13 May/June 2022 · Question 9

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Question	Answer	Marks
9(a)	Four from: The shell contains the explanation system, inference engine, user interface and knowledge base editor (must have 3) The knowledge base editor/knowledge base editing software enables the knowledge engineer uses to edit the rules and facts within the knowledge base Explanation system explains to a user the chain of reasoning used to arrive /why they arrived at a particular conclusion The user interface is how the computer interacts with the user/ allows the user to interact with the expert system The inference engine is the reasoning/decision-making part of the system/ provides the reasoning of an expert system	4

Question	Answer	Marks
9(b)	Four from: Details of the individual would be entered into the user interface... ...such as the applicant's age, gender, whether they smoke and general health The user interface will ask questions (of the consultant/user) The inference engine will compare the responses... ...with the facts in the knowledge base... ...using the IF...THEN rules The inference engine will produce a recommendation for how much the applicant will pay each month for their policy... ...or recommend whether the application is rejected May inform them of the details of the policy The recommendation is output through the user interface The consultant will use the expert system's recommendations to make a final decision	4

Question	Answer	Marks
5(a)	Four marks available: • An expert system would store the locations of each distribution point (1) • An expert system would store the type of the vehicle being used (1) • An expert system would store the working hours (1) • The scheduler would type in the destinations/quantity of parcels for the day ... (1) ... the expert system would match those against the available types of vehicle (1) ... match against locations (1) • The expert system would suggest how many drivers would be needed (1) • The expert system would suggest the kind of vehicle needed for each driver (1) • The expert system suggests the allocation of orders to each vehicle (1) • The expert system suggests a list of goods in reverse order (1) so that each vehicle had the first order loaded on to the vehicle last (1) • The expert system would take account of (e.g.) fuel and time//traffic and road conditions (1) • Outputs the most efficient route (1)	4
5(b)	Three marks available: • Backward chaining starts with a list of goals/hypothesis ... (1) ... and tries to find support for their hypothesis (1) • An inference engine would search the inference rules until it finds one which has a THEN part that matches the goal/hypothesis (1) • If the IF part of that rule is known to be true, then it is added to the list of goals (1) • This method is called goal-driven (1)	3

Mark scheme · 9626/12 May/June 2023 · Question 9

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Question	Answer	Marks
9	Eight from: • Identification of three of: knowledge base, explanation system, knowledge base editor, inference engine, user interface, rules base (1) • Inference engine is able to find possible diagnoses by using a form of reasoning (1) • The inference engine uses the data or facts in the knowledge base to reason through the symptoms (1) • (User interface/expert) system asks questions/for symptoms about the illness (1) • Answers//symptoms/required data are typed in (1) • Inference engine compares symptoms to those in the knowledge base (1) • Any awareness of use of IF...THEN... rules/comparisons//rules base consists of IF...THEN rules... (If and THEN both used in the answer) (1) • Searches the inference rules to find one where the IF statement is known to be true (1) • If answer not found, further questions are asked (1) • When such a rule is found, the inference engine uses the 'THEN' part ... (1) ... to find the illness (1) • Possible/suggested/a range of diagnoses are output (to user interface) (1) • Explanation system produces reasons for suggestions/diagnoses (1) • Reasons for suggestions output using the user interface (1) • Doctor can choose to use or reject answer (1)	8

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Question	Answer	Marks
10(a)	Forward chaining	1
10(b)	Four from: • An inference engine searched the rules until it found one where the IF statement was true (1) • An inference engine rejects any rule that is not TRUE (1) • The IF Hortense is an Ostrich is NOT TRUE (1) • The IF Hortense sings and Hortense is 15 cm is TRUE (1) • The 'THEN' part caused the addition of new information (1) • Inference engine looked for another matching IF... (1) • The IF X is a canary - THEN X is yellow is TRUE (1) • Inference engine stopped because the goal was reached (1)	4

Question	Answer	Marks
8	Four from: • One mark for identification of all of: Rule Base, Knowledge Base, Inference Engine (1) • Uses forward chaining to find a solution (1) • Makes strategic decisions based on data analysis and interpretation (1). • The data entered into the system determines which rules are selected and used (1) • The inference engine/process starts with the facts/data of a problem and progresses to a goal . • Any explicit awareness of IFTHEN being part of the process (1) • An inference engine searches the rule base (1st) • until it finds one where the IF statement is known to be true (1) • When a suitable rule is found, the inference engine uses the 'THEN' part to cause the addition of new information/facts (to the knowledge base) (1) • This process repeats until a goal is reached (1)	4

Question	Answer	Marks
5(a)	Four from: • A user interface so questions/queries about (e.g.) financial details appear (1) • A user interface so answers to questions can be input (1) • Knowledge base editor to enter/amend (e.g.) current financial facts and figures (1) • Database of facts/knowledge base to hold/store current financial details (1) • Rules base to hold/store financial rules, (1) • Inference engine to interrogate knowledge base//uses forward/backward chaining to come to a conclusion (1) • Explanation system to describe why/say why/justify how the conclusions were arrived at. (Do not award “explains”) (1)	4
5(b)	Six from: • Reduced staff costs (1) • Saves staff time in writing (e.g.) documentation (1) • Makes bank more competitive (1) • Expert systems give better advice than humans (1) • Increase customer satisfaction (1) • Makes the bank look more reliable (1) • Expert system, does not need e.g. breaks/holidays (1) • Cheaper to copy an expert system than train new staff (1) • Errors are reduced. (1)	6
5(c)	Three from: • It can lead to bad reviews/reputation (1) • It can lead to loss of customers (1) • Expert systems cannot apply common sense (1) • Expert systems cannot be creative in//cope with unusual situations (1) • Costs of creation//initial installation//initial purchase is high (1) • Knowledge base/rules require frequent updating//knowledge goes out of date (1) • Need to pay/employ a (e.g.) technician//pay maintenance costs for the expert system (1)	3

Question	Answer	Marks
3	Six from: Forward chaining MAX FIVE - Forward chaining starts with/uses the data (1) - Uses inference rules to extract more data. (1st) - until a goal is reached (1) - An inference engine searches until it finds an IF statement known to be true (1) - The inference engine uses the 'THEN' part of the IF THEN statement to cause the addition of new information (1) - Inference engines will iterate through this process (1st) - until a goal is reached (1) - This method is called data-driven (1) Backward chaining MAX FIVE - Backward chaining starts with/uses a list of goals/hypotheses (1) - Uses inference engine to search for data that fits a goal/hypothesis (1) - An inference engine searches until it finds a THEN part that matches a desired goal (1) - If the IF part of that rule is not known to be true (1st) - then it is added to the list of goals (1) - This method is called goal-driven (1)	6

Question	Answer	Marks
10	EIGHT from: Max five if only first points Knowledge base comprises a database of facts and a rules base (1st) - It is the information used to make the decision (1) - Database of facts gathered from experts (1) Rules base is a set of rules/statements (1st) - usually of the form IF . . . THEN//description of IF ...THEN in practice in context (outcomes based on what is seen/observed/measured) (1) The shell includes the user interface, explanation system, inference engine and knowledge base editor (THREE FROM) (1) Inference engine is the reasoning part of the system (1st) - That uses forward chaining or backward chaining (1) User interface is how the computer interacts with the user, (1st) - displays questions/information (1) - enables the user to type in answers to the questions (1) Knowledge base editing software/knowledge base editor enables the knowledge engineer to edit rules and facts (1) Explanation system explains to a user the chain of reasoning used to arrive at a particular conclusion. (1)	8

Mark scheme · 9626/12 Oct/Nov 2024 · Question 10

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Question	Answer	Marks
10	FOUR from: • Contains the knowledge of many experts/zoologists/scientists (1) • Produce results quicker//reduces time taken to produce results (1) • No need to carry/have access to paper records/ information/ reference books/remember a great deal of information (1) • Outcome is consistent (1) • Fewer errors in identification of animal species (1) • Can enhance scientist's/zoologist's own knowledge (1) • Provide permanent record of identification of animal (1) • No need to check with other experts/zoologists/scientists if in doubt over findings. (1) • Expert systems provide justifications/explanations for decisions (1)	4

Mark scheme · 9626/13 Oct/Nov 2024 · Question 7

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Question	Answer	Marks
7	FOUR from: • It is used in backward chaining (1) • It starts with a goal/set of goals (1st) – which establish the rules to be followed (1) – to identify initial conditions (1) • Inference engine uses the rules base (1) – to find one which has a THEN part that matches a required goal (1) (Award once only) • Inference engine explores the rules (1st) – until it finds one which has a THEN part that matches a required goal (1) (Award once only) • If the IF part of that rule is known to be true (1st) – it is added to the list of goals. (1)	4

Question	Answer	Marks
4(a)	One mark per bullet point to a maximum of three marks. • An expert system is able to consider various factors/inputs (1st) – such as distance/traffic/driver workload/customer preference (1) • It is a more efficient method of solving the task. (efficiency in production or of outcome – accept either, but , production of outcome OR quality of outcome must be stated) (1) • Fewer scheduling/human errors (1st) – such as forgetting about time windows/overlooking optimal routes/drive availability (1) • The company will be able to keep their scheduling system active 24/7 (1) • Reduced expenses/fuel usage/wear and tear (1). • Chooses the quickest/most fuel-efficient route (1) • The company will be able to scale their operations//An expert system will allow for improved business scalability/no need to employ new staff as business grows (1) • Able to produce a picking order (1) • Deliveries can be loaded in reverse order of delivery (award concept of) (1)	3

Question	Answer	Marks
4(b)	One mark per bullet point to a maximum of six marks. • <u>Gather information</u> about delivery processes. (1) • TWO marks for items of information that would be gathered (e.g.) – Distances (1) – Locations (1) – Speed limits (1) – Capacity of individual vehicles (1) – Road capacities (e.g. low bridge clearances) (1) • Consult with expert schedulers to understand decision-making processes/rules followed/schedule trade-offs. (1) • Decide how to encode/design/create//gather the acquired knowledge (1) turning it into a <u>knowledge base</u>. (1) • Define/create <u>rules base</u> (1) • Any example of a rule that may be in the rules base (1) • Design/create a <u>user interface</u> (1) that allows schedulers/users to interact with the system (1) • Create the inference engine (1) • Create the explanation system (1) • Test the <u>system</u> with archived data/real-world data/simulated scenarios (1) and improve as necessary (1)	6

Mark scheme · 9626/12 May/June 2025 · Question 3

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Question	Answer	Marks
3(a)	One mark per bullet point to a maximum of two marks • Questions are displayed on the user interface (1) • The user types in the problems/answers the questions (1) • Outputs explanation of findings (1) • Outputs suggestions/probabilities of possible solutions. (1)	2
3(b)	One mark per bullet point to a maximum of two marks. • Acts as the reasoning system (1) • Compares the faults with those held in the knowledge base (1) • Uses/searches the rules base//set of rules (1) • Uses IF THEN (1) • Use backwards/forwards chaining (1) • Produces suggestions/solutions (1).	2

Mark scheme · 9626/13 May/June 2025 · Question 3

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Question	Answer	Marks
3	One mark per bullet point to a maximum of six marks. • Inference engine compares inputs with rules in the knowledge base (1) • Uses IFTHEN reasoning (1) Forward chaining • The reasoning uses forward chaining (1) • Forward chaining starts with the available data (1st) – And uses inference rules to extract more data//ask more questions (1) – Until a goal is reached//Until it finds one where the IF statement is known to be true (1) – The THEN part adds new information (1) • Forward chaining can be called data-driven (1) Backward chaining • The reasoning uses backward chaining (1) • Backward chaining starts with a list of goals/hypotheses (1) • Backward chaining would search the inference rules (1st) – Until it finds one which has a THEN part that matches a desired goal (1) • If the IF part of that rule is not known to be true (1st) – The THEN part adds this to the list of goals (1) • Backward chaining can be called goal-driven (1)	6

Question	Answer	Marks
4(a)	One from: A database/store of facts and a rules base (1)	1
4(b)	Four from: - Knowledge base holds facts (1) gathered from an expert (1) - Knowledge base holds rules (1) that are used to interrogate the facts (1) - It is interrogated by the inference engine (1st) - Using relevant rules//rules that apply (1) - The rules base includes IF...THEN rules (1)	4

Question	Answer	Marks
3(a)	Four from: - Acts as a reasoning system/applies logic/makes decisions (1) - Searches the knowledge base/database (1) - Selects/applies/uses applicable/relevant rules (1) - Uses rules to interrogate the knowledge base/database (1) - Uses the rules to determine outcomes (1) - Uses IF THEN analysis (1) - Uses forward chaining/backward chaining to reason/arrive at solutions (1) Request clarification from the user if necessary/asks (further) questions from the user if necessary (1)	4
3(b)	Four from: - Data is collected from experts/relevant sources/data mining (any awareness of source) (1) Relevance of data is decided (1) - Data is entered (into knowledge base) (1) - Uses a knowledge base editor to edit/created by a knowledge base engineer (1) - Design/decide the rules base (1) Create the rules base (1) - Create the links between the rules and knowledge/data (1)	4

Question	Answer	Marks
6	FIVE marks available: • Use of an example (1) FOUR from: • It is part of/uses forward chaining (1) • A process of using/analysing data to find a solution (this mark may be awarded for a description of this process as part of the description of the context) (1) • Data entered into the system determines the rules to be used (1) • (The inference engine) starts with the data and progresses towards a goal (1) • (The inference engine) will try to match the (inference) rules//searches the rules (award any indication that the rules base is interrogated) (1st) Until one matches the IF statement (1) • The THEN part of the rule brings out new information/data/advice (may answer in context instead of saying information/data/advice) (1) • Process is repeated until a goal is reached//no further possible solutions can be found (1)	5

Question	Answer	Marks	Guidance
2	EIGHT marks available: • A user interface is used to input (1) • IF....THEN (any awareness of USE of) (1) • Any awareness of the use of rules (may exemplify) (1st) These rules are stored in the knowledge base (1) • An expert system takes prior learning/knowledge and applies it to a problem (1) • Locations/address (for delivery) are input (1) • Details of delivery are input//examples of details (weight/dimensions etc) (1) • The responses will be compared with the facts (1st) By the inference engine (1) In the knowledge base (1) • Suggested routes (1) and estimated times of delivery will be output (1) To the user interface//to a printed list (1)	8	Do Not Award • One word answers – must be the USE of. • ANY answers that are not in the scenario • GPS (references to real time processing, traffic delays etc may be taken as GPS) • Parcel collection The question states ‘in this scenario’ therefore, answers have to be in this scenario

Question	Answer	Marks	Guidance
7(a)	TWO from: - Knowledge base editor (1st) allows the engineer to update the knowledge base (1) allows the engineer to edit the rules base (1) - Knowledge base (1st) stores all the facts/information (1) stores the rules base (1) - Inference engine (1st) Applies logic (1) Uses IF THEN reasoning (1) (Uses reasoning to) identify possible solutions/conclusion (1) Uses forward chaining/backward chaining (1) - Rules base (1st) Contains the IFTHEN reasoning (that is relevant to the expert system) (1) - Contains the FORWARD/BACKWARD chaining to reach a solution/conclusion (1) Contains the rules that the expert system uses to draw conclusions (1)	4	Do Not Accept: - User interface systems - Explanation systems

Question	Answer	Marks	Guidance
7(b)	EIGHT marks available: The final decision to follow the advice given by the expert system rests with the mechanic (this may be presented as a positive or a negative aspect) (1) Benefits Max 6: e.g. - Provides a consistent analysis (1) - Knowledge is based on input from a range of experts (1st) therefore is based on people who have solved similar issues (1) ONE FROM as (1st) Uses a process of reasoning based on facts and rules Uses what-if analysis Uses IF THEN analysis so there is a structured process of reasoning (1) - Produces a range of possible solutions (awareness of more than one) (1st) based on answers to a set of questions (1) - Provides full coverage//covers all areas/issues (1) - Includes areas about which the mechanic does not know a great deal (1) - Provides an explanation for decisions (1st) therefore the mechanic can learn (1) - Available at all times (1st) therefore do not need to contact another mechanic at an inconvenient time (1) Drawbacks Max 6: e.g. - Can only deal with problems that are recorded in the expert system (1st) therefore anything outside of the scope/experience of the experts/knowledge base cannot be dealt with (1) - May need to create a new expert system//update current expert system (1) As new cars/engines come onto the market//new technology is used in cars (1)	8	

Question	Answer	Marks	Guidance
7(b)	- If <u>questions are badly structured</u>, will not have all the information needed to solve the problem (1st) therefore suggested solutions may be irrelevant/damage the car further (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) - One mark for a value judgement/conclusion after the discussion of at least two arguments (1)		

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Question	Answer	Marks	Guidance
10(a)	TWO from: - Uses IF THEN reasoning (1) - Uses forward/backward chaining (1) - Applies the rules to the data//interrogates the knowledge base (1) - To identify possible solutions/answers/results (1) - Application to identification of flowers (must be more than mentioning flowers and must be more than a repeat of question) (1)	2	
10(b)	TWO from: Holds all the facts (1) such as named item from identification of flowers (1) Holds all the rules (1) Holds all the heuristics (1)	2	